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PostgreSQL - Tuning Database-Level Parameters and Defaults for Optimal Performance — PostgreSQL Foundations Series

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PostgreSQL is robust out-of-the-box, but default configurations are often conservative to support wide compatibility. In production—especially with modern hardware and varying workloads—tuning is essential for performance, stability, and scaling.

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2. Key Principles of PostgreSQL Tuning

- **Understand the workload**: OLTP vs OLAP?
- **Measure before tuning**: Always benchmark with pg_stat_statements, EXPLAIN, or pgbench.
- **Tune iteratively**: Small adjustments → test → monitor.

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3. Resource-Aware Defaults: Tuning for Your Environment
Example Baseline by Memory Size

Memory	`shared_buffers`	`work_mem`	`maintenance_work_mem`
4 GB	1 GB	4 MB	64 MB
16 GB	4 GB	16 MB	256 MB
64 GB	16 GB	64 MB	1 GB

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4. Important Database-Level Parameters
A. Memory Settings

- shared_buffers: Memory allocated to PostgreSQL buffer cache. (25–40% of system RAM)
- work_mem: Used per sort/join operation. Too high → risk of out-of-memory.
- maintenance_work_mem: Used for vacuuming, creating indexes.

B. Checkpoint Settings

- checkpoint_completion_target: Set to 0.9 for smoother IO.
- checkpoint_timeout: Increase for large write-heavy systems.
- wal_buffers: Set to 16MB or -1 (auto).

C. WAL Settings

- wal_level: Use replica or logical depending on replication needs.
- max_wal_size / min_wal_size: Increase to avoid frequent checkpoints.
- wal_compression: Enable for space saving.

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5. Autovacuum and Housekeeping

Autovacuum is essential. Tuning it prevents bloat and improves indexing.

- autovacuum_vacuum_threshold: Reduce for large tables.
- autovacuum_vacuum_cost_limit: Raise this for aggressive cleanup.
- log_autovacuum_min_duration: Log vacuum times for analysis.

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6. Query Optimizer and Planner Control

- random_page_cost: Lower for SSDs (e.g., 1.1 instead of 4).
- effective_cache_size: Estimate of OS-level disk cache; ~70–80% of system memory.
- default_statistics_target: Increase for better planner choices.

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7. Connection Settings

- max_connections: Too high leads to memory issues. Use a pooler like PgBouncer.
- superuser_reserved_connections: Reserve access for admin/monitoring.

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8. Logging for Diagnostics

- log_min_duration_statement: Set for slow query detection.
- log_line_prefix: Include user, DB, PID, duration for traceability.
- log_checkpoints, log_lock_waits: Enable to surface bottlenecks.

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9. Useful Extensions for Tuning Insight

- pg_stat_statements: Top queries and time spent.
- pg_stat_kcache: OS-level resource usage.
- pg_buffercache: Check buffer usage patterns.

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10. Cloud-Specific Tuning Tips

- **AWS RDS/Aurora**: Some parameters require parameter group changes.
- **GCP Cloud SQL**: Limited tuning surface; use query optimization first.
- **Azure Database for PostgreSQL**: Tuning via portal or CLI; latency-sensitive workloads benefit from random_page_cost tuning.

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11. Example Tuned Configuration for 16GB Host

```
# Memory
shared_buffers = 4GB
work_mem = 16MB
maintenance_work_mem = 256MB
effective_cache_size = 12GB

# WAL
wal_level = replica
wal_compression = on
checkpoint_completion_target = 0.9

# Autovacuum
autovacuum_vacuum_threshold = 50
autovacuum_vacuum_cost_limit = 2000
log_autovacuum_min_duration = 0

# Planner
random_page_cost = 1.1
default_statistics_target = 100

# Connections
max_connections = 100
superuser_reserved_connections = 3

# Logging
log_min_duration_statement = 1000
log_checkpoints = on
log_line_prefix = '%t [%p]: [%l-1] user=%u,db=%d '
```

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12. Monitoring and Verification

Always monitor after tuning using:

- pg_stat_statements
- pg_stat_bgwriter
- pg_stat_activity
- pg_stat_user_tables

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13. Automating Tuning With Tools

- pgtune: Auto-generate based on hardware specs.
- pganalyze, pgmetrics: Visual dashboards.

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14. Common Pitfalls

- Over-allocating work_mem
- Not adjusting for storage type (SSD vs HDD)
- Ignoring autovacuum settings
- Blindly copying configs without context

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15. Tuning Is an Ongoing Process

PostgreSQL tuning is not a one-time task. As your workload evolves, revisit your configuration.

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